

Replication

Step 1: Prepare the Repository

Clone or copy the repository into a clean working directory, then verify that the CSV datasets exist in data/. The repository already includes the required Lab 3 files.

Step 2: Install Dependencies

Command

```
python -m pip install -r requirements.txt
```

Step 3: Run the Final Experiments

Experiment command

```
python run_experiments.py \  
  --systems 7z LLVM x264 \  
  --budgets 10 20 30 40 50 \  
  --repeats 30 \  
  --seed 20260424 \  
  --initial-design 6 \  
  --candidate-pool 4096 \  
  --kappa 0.5 \  
  --explore-prob 0.1 \  
  --output-dir outputs/final_run
```

Step 4: Rebuild Figures, Tables, and PDFs

Command

```
python build_submission_docs.py --artifact-link "https://github.com/zhaibuqun-cell/rf-smbo-configuration-tuning"
```

Expected Outputs

- outputs/final_run/raw_results.csv
- outputs/final_run/summary_results.csv
- outputs/final_run/statistics.csv
- outputs/final_run/figure_1_convergence.png
- outputs/final_run/figure_2_per_system.png
- coursework_report.pdf, requirements.pdf, manual.pdf, replication.pdf

Reproducibility Notes

The base random seed is 20260424. Matched seeds are used so that each Random Search run is paired with an RF-SMBO run under the same system and repeat index.

Optimum and worst values are used only after search to compute the normalised gap. They are not consulted during candidate selection.

At budget 50, the expected overall mean normalised gap is 0.0487 for Random Search and 0.0113 for RF-SMBO. Expected headline p-values are $9.31\text{e-}10$ for LLVM and $9.10\text{e-}07$ for x264.